

BIOL10.210 - Objectives for Practical #3 (online A&P I)

Chapter 10

1. label connective tissue layers and structure of a muscle; Figures 10.1 & 10.2
 endomysium, perimysium, epimysium
 fascicle
 muscle fiber
 myofibril
 sarcoplasm
 sarcoplasmic reticulum
 sarcolemma
 T tubule
 terminal cisterns
 mitochondria
2. describe microscopic anatomy of skeletal muscle; Figures 10.3 & 10.4
 actin, myosin, troponin, tropomyosin, titin
 Z disc
 M line
 A, H, and I bands
 sarcomere

Chapter 11

1. identify 26 muscles (refer to “muscle sheet”) on images (Figures throughout chapter); and provide action, origin, insertion, and innervation

Chapter 12

1. identify important anatomical structures of a typical neuron; Figure 12.2
 dendrites myelin sheath
 cell body neurolemma
 Nissl bodies myelin sheath gap/node of Ranvier
 neurofibrils axon
 axon hillock axon terminal
 axon collateral synaptic end bulb
2. identify neuroglia/glia cells and their functions; Figure 12.6 & 12.7

Chapter 13

1. describe anatomy of the spinal cord; Figure 13.1 & 13.3
 white columns – posterior, anterior, and lateral funiculi
 dorsal root ganglia
 dorsal and ventral roots
 spinal nerve
 posterior medial sulcus
 gray matter – posterior, anterior, and lateral horns
 gray commissure
 central canal
 anterior medial fissure
 denticulate ligament
 meninges – dura mater, arachnoid mater, pia mater

2. know connective tissue coverings and structure of a nerve; Figure 13.5
 - endoneurium, perineurium, epineurium
 - fascicle
 - axon
 - myelin sheath
 - spinal nerve

Chapter 14

1. be able to locate/identify external/internal features of the brain; Figures 14.1, 14.8, & 14.11
 - cerebrum - frontal, parietal, temporal, occipital, and insula lobes
 - central sulcus
 - precentral gyrus
 - postcentral gyrus
 - longitudinal fissure
 - transverse fissure
 - lateral sulcus
 - cerebellum – arbor vitae, vermis, folia
 - diencephalon – thalamus, hypothalamus
 - brainstem – pons, medulla oblongata
 - pituitary gland
 - pineal gland
2. identify the 3 meningeal layers and associated structures; Figures 14.2
 - dura (periosteal and meningeal layers), arachnoid, and pia mater
 - subarachnoid space
 - dural sinus
 - arachnoid villi
3. be familiar with CSF formation, circulation in ventricles, and drainage of CSF; Figures 14.3 & 14.4
 - lateral, third, and fourth ventricles
 - lateral and medial apertures
 - cerebral aqueduct
 - choroid plexus
 - interventricular foramen
 - aqueduct of midbrain
 - central canal
4. identify 12 cranial nerves – know name and number; image from powerpoint